

Short term exposure to polystyrene nanoplastics in mice evokes self-regulation of glycolipid metabolism

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ABSTRACT

With the detection of nano-plastics (NPs) in daily essentials and drinking water, the potential harm of NPs to human health has become the focus of global attention. Studies have shown that long term exposure to NPs can lead to disorders of glucose and lipid metabolism in organisms, while the effects of short term exposure are rarely reported. Moreover, environmental factors cause the aging of NPs, and it is unclear whether this has an effect on their toxicity. In this study, we use 100 nm polystyrene (PS) NPs and ultraviolet (UV) aging PS (aPS) NPs to gavage mice for 7 days at an exposure dose of 50 mg/kg/day. To evaluate the effects of exposure on mice hepatic glucose lipid metabolism, we performed blood biochemical, pathological and metabolomic analyses. The results showed that exposure to PS NPs and aPS NPs increased serum glucose, disrupted serum lipoprotein levels, and up-regulated the expression levels of phosphatidylinositol 3-kinase (PI3K)/ phosphoprotein kinase B (p-AKT)/ Glucose transporter 4 (GLUT4) proteins in the glucose metabolism pathway. The expression levels of key proteins sterol regulatory element binding protein-1 (SREBP-1)/peroxisome proliferator-activated receptor- γ (PPAR γ)/adipose triglyceride lipase (ATGL) in the lipid metabolism signaling pathway were significantly increased. These findings suggest that short term exposure to PS NPs and aPS NPs induces glycolipid metabolism disturbance in mice, which may subsequently awaken the mice to self-regulate the serum levels of various lipoproteins and the expression of related key proteins. Compared with PS NPs, the aPS NPs interfered more strongly with glucose metabolism, and the corresponding self-regulation in mice was also more obvious. These findings not only provide a basis for environmental factors to increase the health risk of NPs but also provided a reference for the selection of test substances for further studies on the toxicity of NPs.

1. Introduction

Plastics can be degraded into plastic fragments under the combined action of long term sunlight, oxidation, hydrolysis, and the decomposition of soil microorganisms in the environment. The US National Oceanic and Atmospheric Administration defines plastic fragments smaller than 5 mm as microplastics (Ter Halle et al., 2017), and microplastics can be further degraded into NPs smaller than 1 μ m (Dawson et al., 2018). Recent research shows that ordinary single-use consumer plastic products release trillions of nanometers or less per liter of NPs into the water during normal use (Zangmeister et al., 2022). As an emerging nano-pollutant, NPs can enter the human body through multiple channels such as the respiratory tract, digestive tract, and skin contact. Studies have shown that NPs can cause lipid metabolism and

behavioral disorders in organisms, and can even penetrate the blood–brain and placental barriers, resulting in neurotoxicity and embryotoxicity, and affecting the development of offspring (Bouwmeester et al., 2015; Brown et al., 2001; Revel et al., 2018). PS is one of the most widely used industrial materials, and the PS NPs derived from it have become a common pollutant in the ocean, soil, and atmosphere (Zhang et al., 2017). PS NPs are reported to have toxicological effects associated with oxidative stress, inflammation, mitochondrial dysfunction, lysosomal dysfunction, endoplasmic reticulum stress, apoptosis (Heddaagard and Møller, 2020; Schirizzi et al., 2017; Shao et al., 2019). Therefore, the potential harm of PS NPs to human health has attracted more and more attention worldwide. However, most of the current studies use engineered monodisperse PS NPs, which are not representative of the PS NPs present in real environments accurately. The oxygen-containing

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functional groups on the aged-NPs surface increased, and the morphology, structure and properties will change significantly, thus affecting its environmental behavior and increasing ecological risks (Hüffer et al., 2018; Kalčíková et al., 2020; Liu et al., 2019b), but their toxic effects and mechanisms on mammals are rarely reported.

As one of the main metabolic organs in the body, the liver can not only bioaccumulate, metabolize, and detoxify foreign substances but is also the target of many pollutants and plays an important role in regulating glucose and lipid metabolism. Studies have shown that PS NPs cause disturbances in lipid and energy metabolism in zebrafish livers and lead to dramatic changes in their locomotion, aggression, and predator avoidance behavior (Sarasamma et al., 2020). Long term exposure to PS NPs can also lead to disturbances in hepatic glucose metabolism in mice and eventually leads to diabetes (Fan et al., 2022). Because the liver is extremely sensitive to foreign poisons, short term exposure may cause changes in its metabolism. However, few studies have reported the effects of short term exposure on hepatic metabolism in mice.

To gain a more objective and in-depth understanding of the health risks of PS NPs, this study took PS NPs and aPS NPs as the research objects, to study their effects on liver glucose and lipid metabolism in mice. By combining metabolomics, biochemical and histopathological examinations, as well as changes in the hepatic PI3K/AKT/GLUT4 signaling pathway and SREBP-1/PPAR γ /ATGL protein expression, we further explored the regulatory mechanism of PS NPs and aPS NPs on abnormal hepatic glucose and lipid metabolism in mice.

2. Material and methods

2.1. Preparation and characterization of aPS NPs

We obtained aPS NPs by irradiating PS NPs with UV light with reference to previous literature (Liu et al., 2019a). Specifically, 5 mg/ml PS NPs (100 nm) suspension was added to the quartz plate and irradiated continuously for 14 days under UV light (18 W, $\lambda_{\max}$ = 254 nm). The liquid level and optical distance were about 5 cm, and the liquid shook every 12 h to ensure uniform irradiation. The prepared aPS NPs were stored at 4 °C for subsequent experiments. Fourier transform infrared spectroscopy (FT-IR, Thermo Fisher Nicolet iS5, US) was used to determine the chemical bonds and functional groups of the PS NPs and aPS NPs. A laser particle size analyzer (Malvern ZS90, UK) was used to characterize the surface zeta and hydrated particle size distribution of PS NPs, and surface morphology changes of aPS NPs were observed by scanning electron microscope (SEM, Hitachi S-4800, Japan).

2.2. Animals treatment

Male ICR mice (5 weeks old; body weight approximately 20 g) were purchased from SPF Biotech (Beijing, China) and were divided randomly into three groups (PS NPs exposure, aPS NPs exposure, and control groups), with six mice in each group. Each mouse in the exposure group was given continuous gavage of either PS NPs or aPS NPs suspension (50 mg/kg/day), and the control group was given an equal volume of pure water. The rearing temperature was between 20 and 26 °C, and the humidity ranged from 40% to 70%. Preliminary observations of food intake, fur, behavior, mental status, urine, and feces were conducted daily for each mouse. On the 7th day after exposure, the mice in each group were sacrificed by cervical dislocation after eyeball blood collection. After weighing, a small piece of liver tissue was removed from the upper left lobe and fixed in 4% paraformaldehyde solution for 24 h, the rest liver tissue was stored in the - 80 °C refrigerator. The feeding of mouse and related experimental operations met the relevant requirements of the Ethics Committee of Capital Medical University (Beijing, China) (ethical review number: 2018-0003).

2.3. Analysis of glucose and lipid in serum and liver

Blood (600 μ L) was collected from the eye, immediately transferred to a sterile centrifuge tube, centrifuged at 3000 rpm/min for 10 min, and then the supernatant was transferred to another centrifuge tube for further analysis. An automatic blood biochemical analyzer was used to detect serum glucose, triglycerides (TG), total cholesterol (TC), high density lipoprotein (HDL-C), and low-density lipoprotein (LDL-C). The levels of liver TC and TG were detected by assay kits (Nanjing, China).

2.4. Histopathology

Liver tissue were first embedded in paraffin wax. Then use a paraffin slicer to cut slices with a thickness of about 4–8 μ m. The cut tissue sections were placed on slides and soaked in warm water at 40 °C to fully extend. Subsequently, the sections were dewaxed in xylene, dehydrated with gradient ethanol, then soaked in hematoxylin and eosin solutions for 5 min, and rinsed with DDW. Then dehydrated with gradient ethanol again, treated with transparent xylene and sealed with neutral glue. Finally, the sections were scanned by automatic scanning system (3DHitech, Hungary), and the liver histological analysis was performed by randomly selecting the field of vision.

2.5. Samples preparation and ^1H NMR measurement

Mouse liver (100 mg) was added to 1 ml of 80% methanol solution and tungsten carbide beads with a diameter of 4 mm to make a homogenate. After centrifugation at 21,130 g for 10 min, the supernatant was taken and placed in a rotary evaporator to dry into a yellow gel-like substance. Then, 500 μ L deuterioxide (D_2O), 30 μ L Phosphate buffer solution (0.1 mM, pH = 7.4), and 20 μ L 0.001% sodium trimethylsilyl propionate were added. The sample was centrifuged at 21,130 g for 10 min at 4 °C. The supernatant (500 μ L) was transferred to a 5 mm diameter NMR tube for subsequent assays.

Nuclear magnetic resonance (NMR) spectrometer (JEOL JES-FA300, Japan) equipped with an ultra-low temperature probe was used to analyze the samples. The specific parameters were set as follows: the proton resonance frequency was 600.13 MHz. The pulse sequence noesygppr1D was used to suppress water peaks (RD-90°-t₁-90°-t_m-90° sampling) and the temperature was set to 298 K. The number of sampling points in the time domain was 32 k, the spectrum width was 12,000 Hz, and the number of accumulations was 128. The change point was 64 k, the exponential line width factor was 1 Hz, and the phase and baseline were corrected with the 3,3,3-trimethylsilylpropionate (TSP) peak center as 0 ppm for chemical shift calibration.

2.6. Data reduction and analysis of ^1H NMR spectra

The Fourier transform, phase correction, baseline correction, and integration of the ^1H NMR spectrum Spectra with MestRenova 5.0 On-line Analysis Website (<https://www.metaboanalyst>). The integration area of the spectrum was 80.9–9.0 parts per million (ppm), and 0.01 ppm is an integration unit. After integration, the integration value obtained was imported into Microsoft Excel 2019, and then into MetaboAnalyst5.0 for principal component analysis (PCA) to obtain the trends of each group. Then orthogonal partial least squares discriminant analysis (OPLS-DA) was performed. To examine the validity of the model, R²Y and Q² parameters were used. According to the variable importance (VIP) in the projection of the OPLS-DA analysis between the two groups, differential metabolites were searched. VIP > 1 and p < 0.05 in univariate analysis were used as the screening criteria for differential metabolite screening.

The metabolite identification used the human metabolome database combined with some metabolomic studies, and the differential metabolite information obtained by screening was input into MetaboAnalyst5.0 for pathway analysis. The impact value > 0.1 and p < 0.05

were used as the standard to screen potential related metabolic pathways.

2.7. Western blot analysis

Liver samples (approximately 50 mg per sample) were weighed and lysed in lysis buffer supplemented with a protease inhibitor cocktail. Total protein concentration was quantified by the bicinchoninic acid (BCA) assay kit (Nanjing, China). Equal amounts of protein were subjected to SDS-PAGE followed by western blotting. Primary antibodies (Abs) were as follows: anti-PI3K Ab (1:1000 dilution), anti-AKT Ab (1:1000 dilution), anti-GLUT4 Ab (1:200 dilution), anti-PPAR γ Ab (1:200 dilution), anti-SREBP-1 Ab (1:200 dilution), anti-ATGL Ab (1:1000 dilution), and anti- β -actin Ab (1:1000 dilution). Goat anti-mouse HRP-conjugated IgG (1:1000 dilution) and goat anti-rabbit HRP-conjugated IgG (1:10000 dilution) were used as secondary Abs. Western blot signals were performed using the Bio-Rad chemiluminescence system (Bio-Rad, US). The bands of the target protein were normalized to the bands of the loading control β -actin. The signal intensity was quantified using the software ImageJ.

2.8. Statistical analysis

Statistical analysis was carried out using SPSS 22.0 software. One-way analysis of variance (ANOVA) and t-test was used to compare differences between groups. The results are expressed as the mean $\pm$ SEM. All of the statistical analyses were two-sided and $p < 0.05$ was considered statistically significant.

3. Results

3.1. Characterization of PS NPs and aPS NPs

SEM images of PS NPs and aPS NPs are shown in Fig. 1a and b. The results showed that the morphology of PS NPs was uniform and spherical, and the surface of aPS NPs was slightly rough, but the morphology did not change significantly. The particle sizes of PS NPs and aPS NPs were 158.8 ± 1.3 nm and 117.0 ± 1.8 nm, respectively. The zeta potentials were -29.6 ± 0.3 mV and -42.4 ± 0.4 mV, respectively. Compared with PS NPs, the hydrated particle size of aPS NPs was reduced, indicating that they are more stable in an aqueous solution. The infrared spectra of PS NPs and aPS NPs are shown in Fig. 1c. Compared with PS NPs, the newly observed absorption peaks on aPS NPs are the OH stretching vibration peaks of hydroperoxides (3740 cm^{-1}), alcohols (3325 cm^{-1}), and carboxylic acids (1772 cm^{-1}), and C=O stretching of

aliphatic and ketones (1712 cm^{-1}). Which is consistent with reports in previous literature (Yousif and Haddad, 2013).

3.2. Clinical chemistry and hematology

No symptoms of illness were observed in the mice at the doses of PS NPs or aPS NPs administered during the experimental period, and all the animals survived until the end of the experimental period. Fig. 2a shows that the blood glucose levels in the PS NPs and aPS NPs exposure groups were significantly higher than those in the control group, and the aPS NPs group was higher than the PS NPs group ($p < 0.05$). Compared with the control group, the serum HDL-C, TG, and TC levels of the exposed group were increased, and the serum LDL-C was decreased. Total cholesterol in both serum and liver was elevated, but TG was significantly lower in the liver and the liver-to-body ratio decreased ($p < 0.05$).

3.3. Histopathology

The histopathological results (Fig. 3) of the liver showed that the hepatocyte structure of the mice in the control group was complete, the nucleus in the center of the cell was clearly visible, and the cords were neatly arranged. In the PS NPs and aPS NPs exposed groups, the hepatocytes were pyknosis, with unclear arrangement of hepatic sinuses, blurred intercellular space, and central hepatic vein congestion. The aPS NPs exposure group showed obvious congestion, accompanied by a large number of vacuoles and hepatocyte balloon-like changes (Fig. 3c).

3.3.1. ^1H NMR spectroscopic and pathway analysis

Thirty-six endogenous metabolites were identified in the NMR spectra of water-soluble liver extract samples (Fig. S1), and Table S1 lists the characteristic groups, chemical shift values, and peak diversity of characteristic peaks for each substance. Fig. 4 shows the resulting score plots for PCA (Fig. 4) and OPLS-DA (Figs. 5 and 6). Each point on the score map (points of the same color represent the same group of samples) represents a sample and the separation of points indicates that there are individual differences between samples. The results showed that PCA could distinguish the normal group from the PS NPs and aPS NPs treated groups without any overlap, the difference between groups was significant. The OPLS-DA model R^2Y and Q^2 of the control group and PS NPs exposed group were 0.986 and 0.958, and the R^2Y and Q^2 of the control group and aPS NPs exposed group were 0.991 and 0.951, indicating high model quality. On combining the VIP value and t-test results, it can be concluded that compared with the control group, β -D glucose, cysteine, cytidine, hypoxanthine, choline, glycine, inosine,

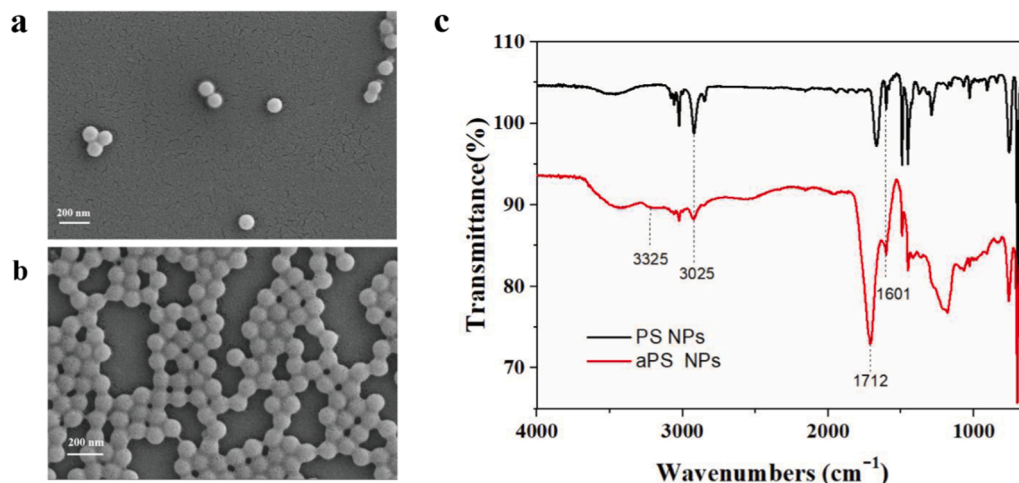


Fig. 1. Characterization of PS NPs and aPS NPs. a. SEM image of PS NPs; b. SEM image of aPS NPs; c. FT-IR spectra.

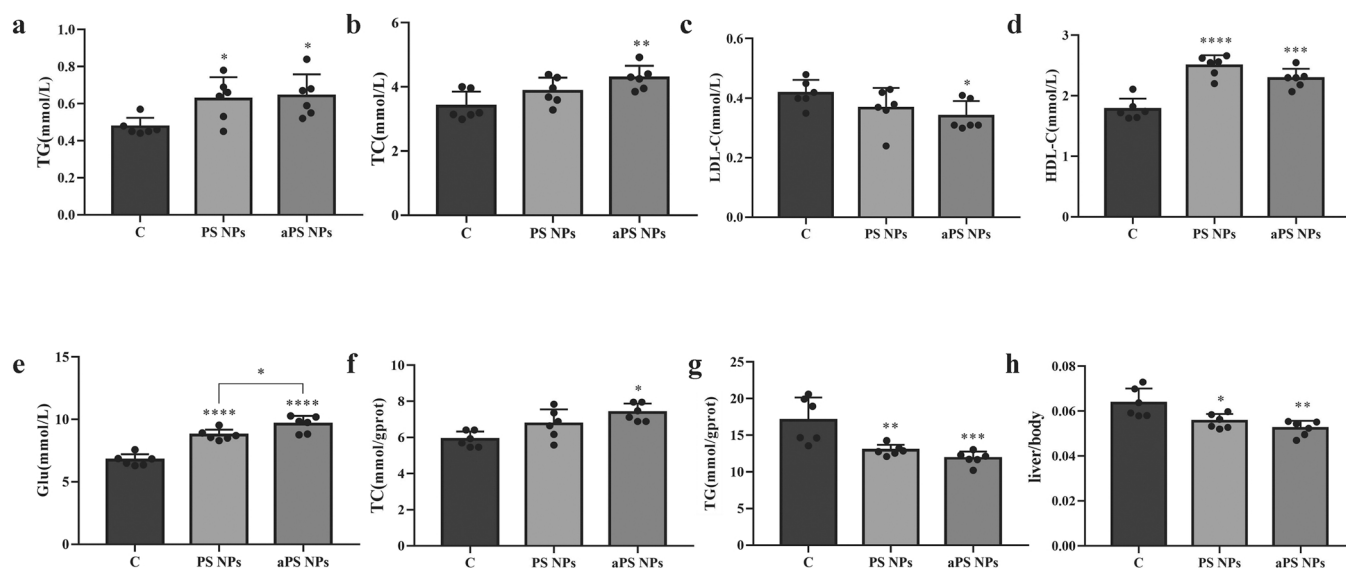


Fig. 2. : Disordered glucose and lipid metabolism in mice after PS NPs and aPS NPs exposure. **a.** serum TG; **b.** serum TC; **c.** serum LDL-C; **d.** serum HDL-C; **e.** blood glucose; **f.** hepatic TC; **g.** hepatic TG; **h.** liver-to-body ratio (n = 6). An asterisk (*) denotes $p < 0.05$, and a double asterisk sign (**) indicates $p < 0.01$, three asterisks (***) indicate $p < 0.001$, and four asterisks (****) indicate $p < 0.0001$.

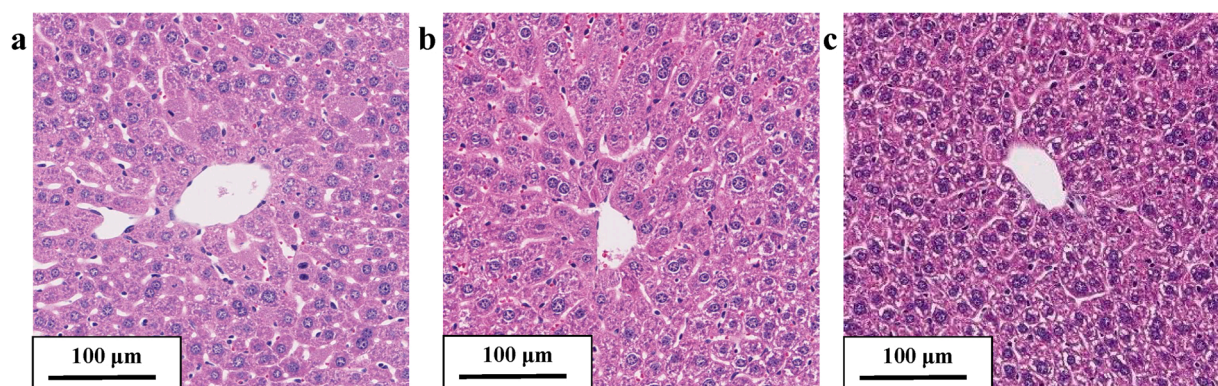


Fig. 3. : Pathological section of liver. **a.** control group; **b.** PS NPs exposure group; **c.** aPS NPs exposure group.

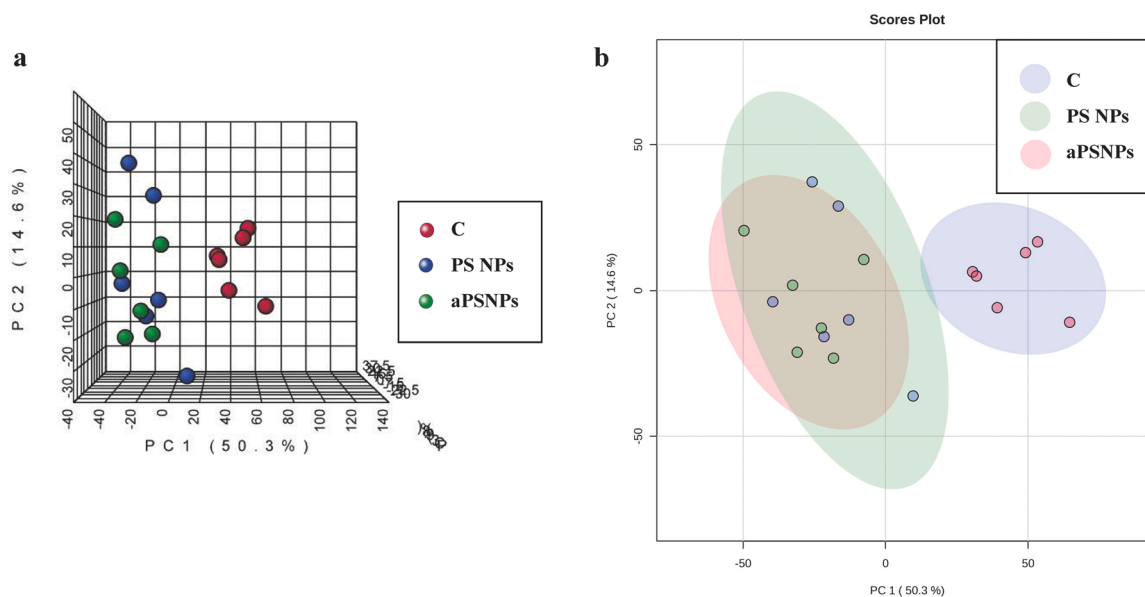


Fig. 4. The synchronized 3D Plots (a) and 2D scores plot (b) of PCA models constructed using ^1H NMR data of livers samples from all groups.

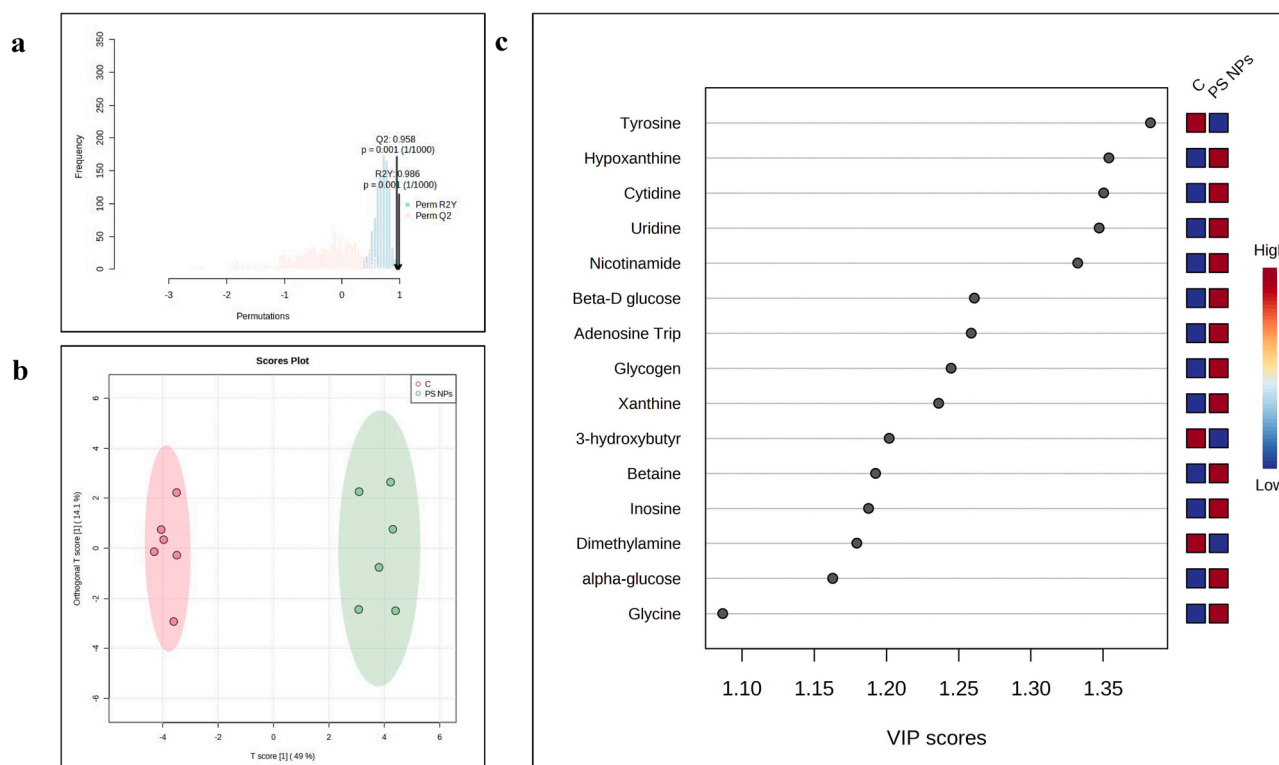


Fig. 5. The model validation (a) and 2D scores plot (b) and VIP-plot (c) of OPLS-DA model constructed using ^1H NMR data of livers samples from control group and PS NPs exposure group.

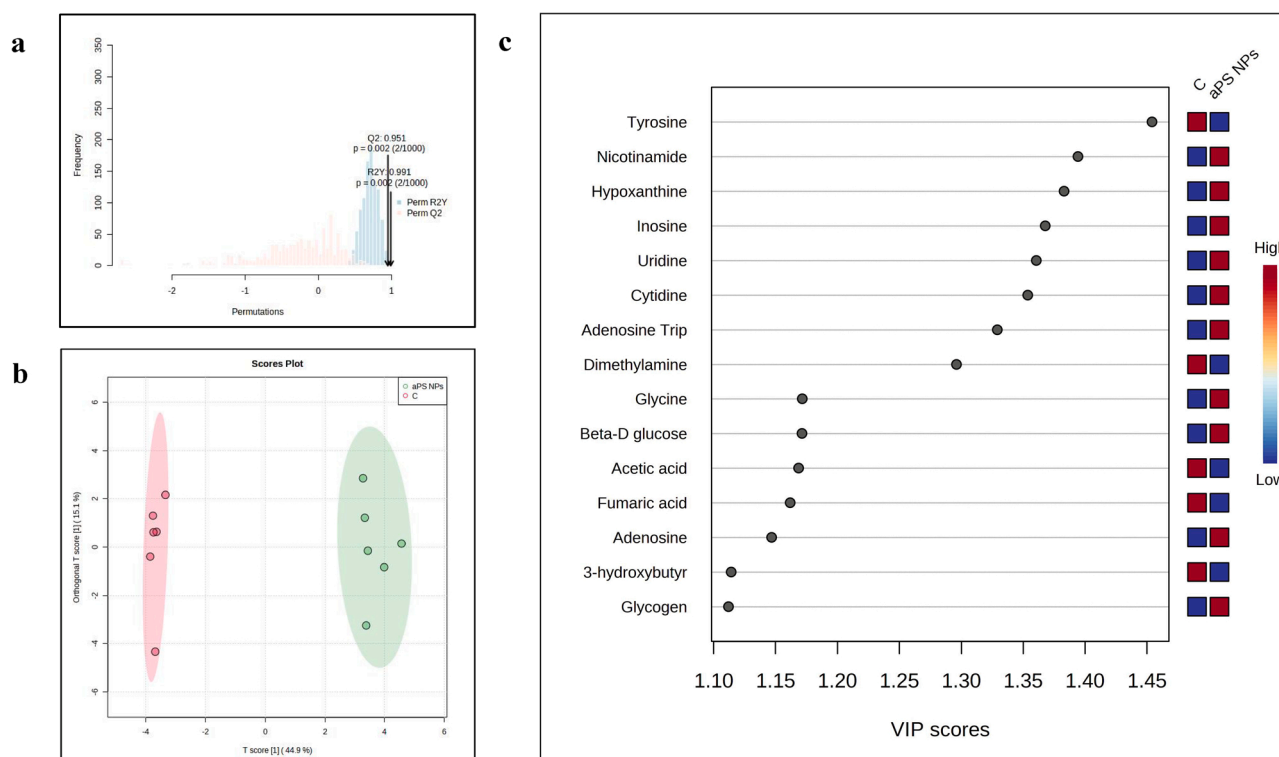


Fig. 6. The model validation (a) and 2D scores plot (b) and VIP-plot (c) of OPLS-DA model constructed using ^1H NMR data of livers samples from control group and aPS NPs exposure group.

uridine, adenosine triphosphate, glycogen, Betaine, and acetate were significantly increased in both PS NPs and aPS NPs groups. In addition, creatine was significantly elevated in the PS NPs group, and 3-

hydroxybutyric acid was significantly decreased. Fumaric acid, succinic acid, adenosine, and nicotinamide were significantly increased in the aPS NPs group, while dimethylamine was significantly decreased.

The metabolites were classified by using a metabolomic online analysis website MetaboAnalyst 5.0, and the relevant metabolic pathways were screened according to the pathway impact value $F > 0.01$, $p < 0.05$. The results showed that PS NPs mainly affected five metabolic pathways, including glycine, serine, and threonine metabolism, purine metabolism, glycolysis/gluconeogenesis, glutathione metabolism, glyoxylate, and dicarboxylic acid metabolism, and pyrimidine metabolism (Fig. S2a). Exposure to aPS NPs mainly affected purine, pyruvate, glyoxylate, and dicarboxylic acid metabolism; glycolysis/gluconeogenesis; the TCA cycle; and the glycine, serine, and threonine metabolism pathways (Fig. S2b). The network diagram related to these metabolic pathways is shown in Fig. 7.

3.3.2. Protein expression of PI3K/p-AKT/GLUT4 and SREBP-1/PPAR γ /ATGL signaling pathways

Compared with the control group, the expression of AKT protein remained unchanged, and the expression levels of PI3K, p-AKT, and GLUT4 in the PS NPs and aPS NPs groups were significantly increased ($p < 0.05$). The expression levels of p-AKT and GLUT4 in the aPS NPs exposure group were higher than in the PS NPs exposure group (Fig. 8). The SREBP-1/PPAR γ /ATGL protein expression in PS NPs and aPS NPs groups was significantly increased ($p < 0.05$), but there was no significant difference between PS NPs and aPS NPs groups (Fig. 9).

4. Discussion

In this study, after 7 days of acute exposure, both PS NPs and aPS NPs groups showed a significant decrease in the liver/body ratio. Pathological sections showed that the nuclei of hepatocytes were pyknotic, the cords were arranged loosely, and there were a large number of vacuoles

in the cytoplasm, indicating that both PS NPs and aPS NPs could cause acute injury to mice liver cells. The disorder in the arrangement of hepatocytes in the aPS NPs exposed group was more obvious, accompanied by a large number of vacuoles and hepatocyte balloon-like changes. A possible reason is that after photoaging, a large amount of environmentally persistent free radicals (EPFRS) and reactive oxygen species (ROS) are formed on the plastic surface (Zhu et al., 2019), and hydroxyl ($\bullet\text{OH}$) and superoxide ($\bullet\text{O}_2^-$) radicals are generated in the process. An important way for nanoparticles to cause cell damage is to induce the production of ROS, increased ROS can lead to oxidative stress in organisms and further damage organelles such as mitochondria (Cui et al., 2021; Liu et al., 2022).

Clinical chemistry and hematology results indicated that exposure to PS NPs and aPS NPs caused a significant decrease in total TG in the liver, but a significant increase in serum TG, which was accompanied by a significant increase in HDL-C levels and a decrease in LDL-C levels. LDL-C helps transport lipids from the liver to the cells of the body and is the primary route for transporting TG and unsaturated fats, and abnormal levels indicate impairments in energy metabolism, lipolysis, and synthesis. Decreased LDL-C and elevated HDL-C levels suggest that the body preferentially stores synthesized lipids in the liver to regulate the reduction of TG in the liver caused by PS NPs and aPS NPs exposure. Both PS NPs and aPS NPs groups increased blood glucose compared with the control group ($p < 0.05$), and aPS NPs had a more significant effect on blood glucose elevation. Metabolomics analysis in this study showed that PS NPs and aPS NPs exposed groups had abnormal levels of small molecules such as creatine, pyruvate, citric acid and creatine, which are closely related to energy metabolism (Philip et al., 2020; Xu et al., 2015). At the same time, acetic acid decreased and liver glycogen increased, suggesting that liver gluconeogenesis was inhibited and liver glycogen

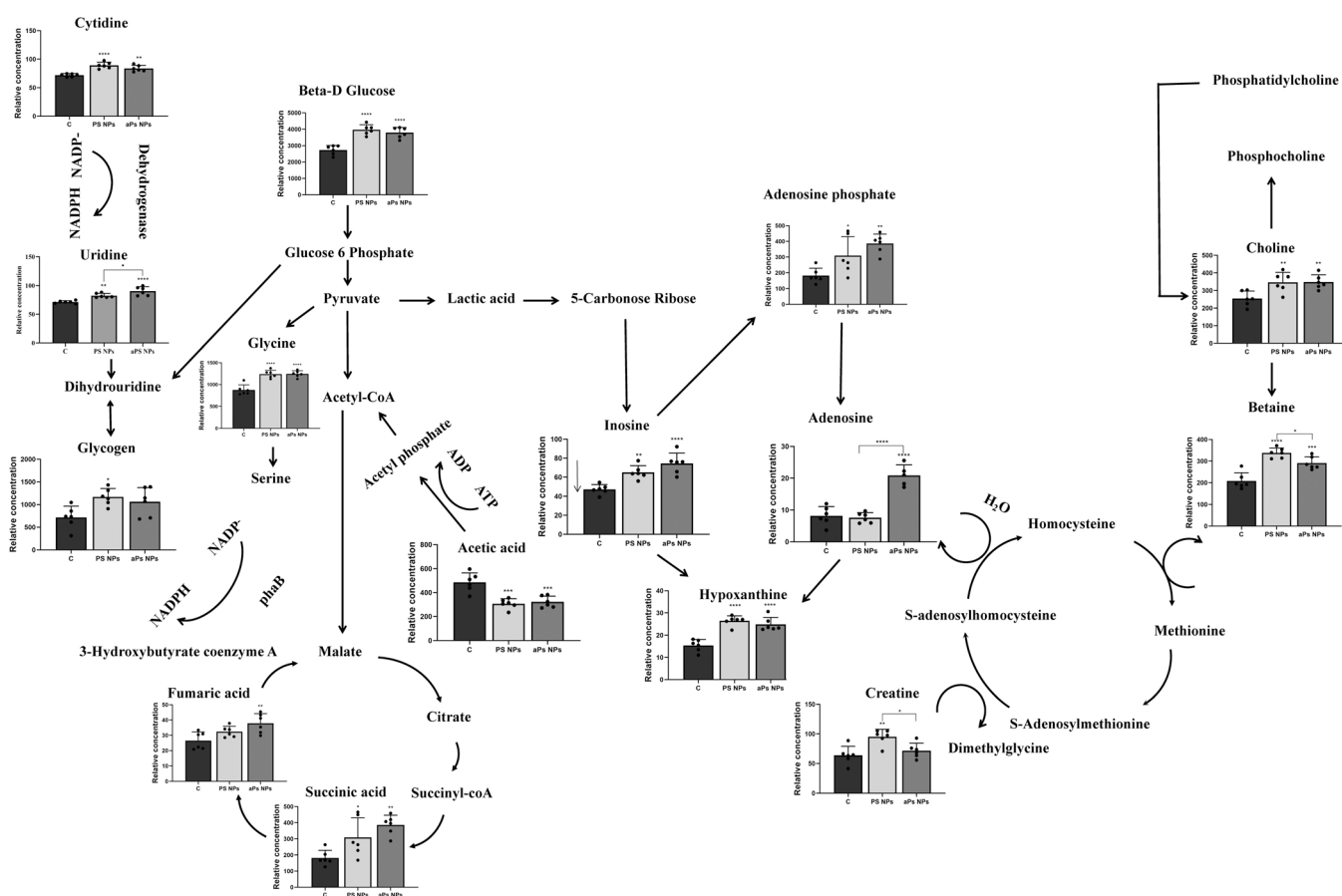


Fig. 7. Overview of the proposed disrupted metabolic pathways for different metabolites detected by ^1H NMR analysis. An asterisk (*) denotes $p < 0.05$, and a double asterisk sign (**) indicates $p < 0.01$, three asterisks (***) indicate $p < 0.001$.

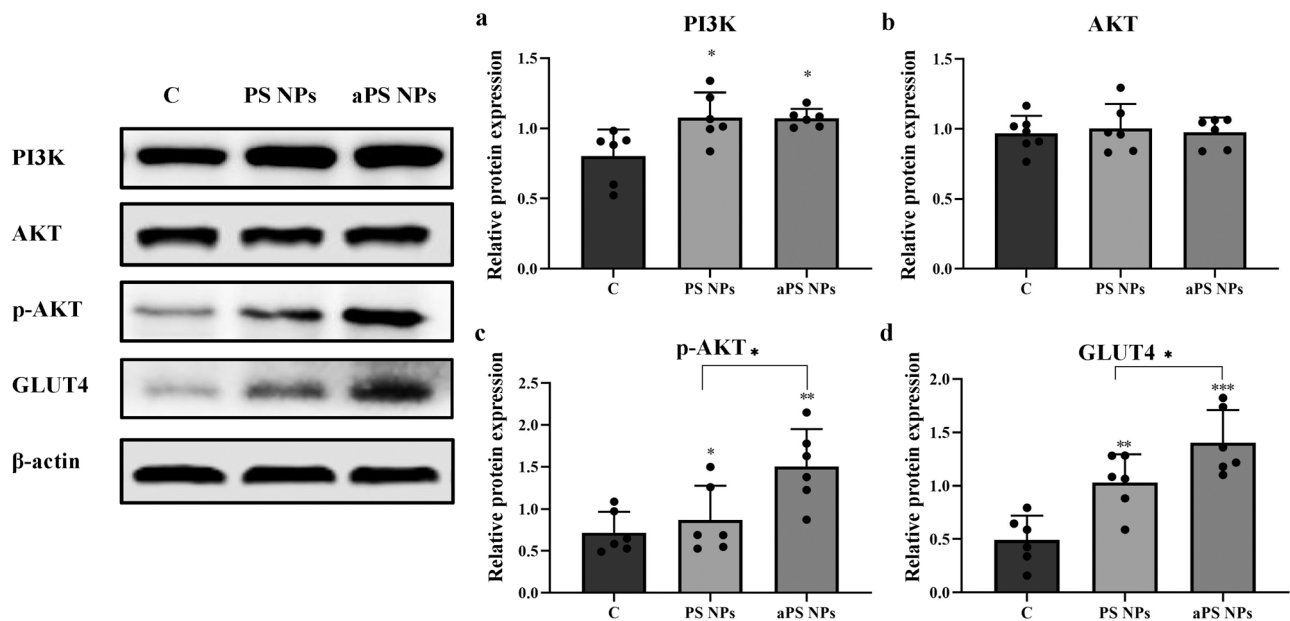


Fig. 8. Protein levels and quantification of PI3K, AKT, p-AKT and GLUT4 in the liver of untreated and PS NPs and aPS NPs treated mice by western blotting (n = 6). An asterisk (*) denotes $p < 0.05$, and a double asterisk sign (**) indicates $p < 0.01$, three asterisks (***) indicate $p < 0.001$.

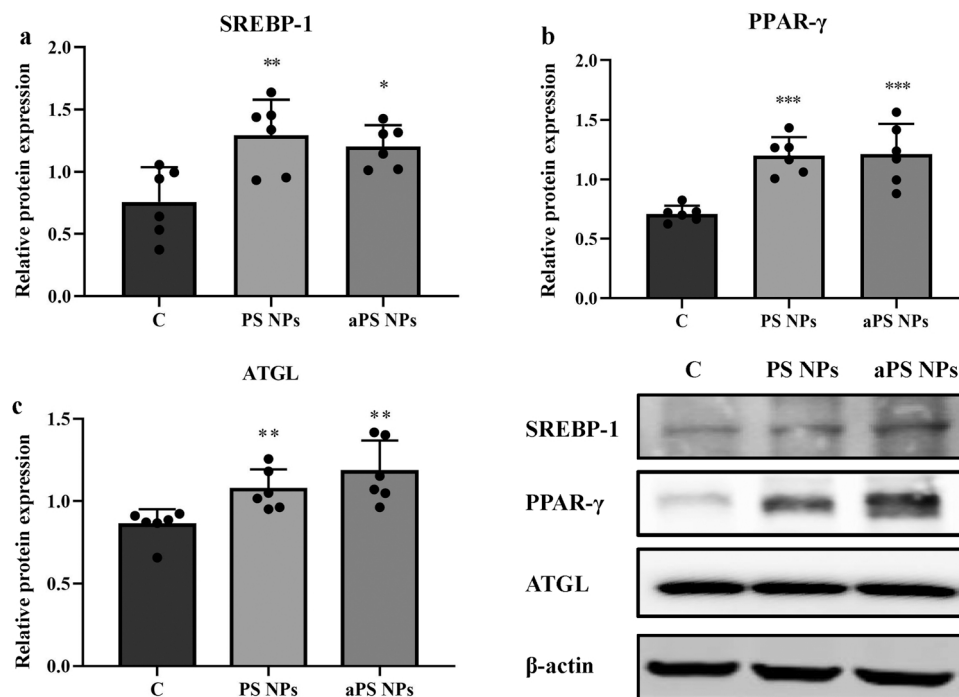


Fig. 9. Protein levels and quantification of SREBP-1/PPAR- γ /ATGL in the liver of untreated and PS NPs and aPS NPs-treated mice by western blotting (n = 6). An asterisk (*) denotes $p < 0.05$, and a double asterisk sign (**) indicates $p < 0.01$, three asterisks (***) indicate $p < 0.001$.

synthesis increased (Chen et al., 2021). Inosine, adenosine monophosphate, hypoxanthine nucleoside and hypoxanthine are important intermediates in purine synthesis, and the increase of their concentration may be caused by local energy supply insufficiency and compensatory change of impaired purine synthesis in liver tissue. These results preliminarily suggested that PS NPs and aPS NPs exposure could disturb glucose and lipid metabolism and energy metabolism in mice. To further explore the related regulatory mechanisms caused by short term exposure of PS NPs and aPS NPs at the protein level, we measured the expression levels of proteins related to glucose and lipid metabolism.

Lipid metabolism is related to a variety of pathway proteins, and abnormal activation or inhibition of lipid metabolism-related pathways can cause lipid homeostasis imbalance. Lipid metabolism is tightly regulated by sterol regulatory element-binding factors, which activate the transcription of genes encoding key enzymes involved in fatty acid and cholesterol biosynthesis (Horton et al., 2002). As a key transcription factor, peroxisome proliferator-activated receptors (PPARs) can selectively upregulate lipogenic enzymes in hepatocytes to enhance lipid synthesis in the liver (Downie et al., 2004). After PPARs bind to ligands and activate, they combine with retinoid X receptor (RXR) to form an

RXR/PPAR γ heterodimer that recognizes and binds to the target gene promoter response element (PPRE) and participates in the regulation of lipid homeostasis (Ji et al., 2013; Suchecka et al., 2011). The main function of ATGL is to catalyze the hydrolysis of the first ester bond of lipid molecules, provide a diglyceride substrate for hormone-sensitive lipase (HSL) hydrolysis, and use triglycerides to provide the necessary energy for organisms (Zimmermann et al., 2004). Western blot results showed that compared with the control group, the expression of ATGL in the liver of PS NPs and aPS NPs exposed mice was significantly increased, which may explain the reduction of TG content in the liver. In contrast, the expression of SREBP-1 and PPAR γ were up-regulated to promote the differentiation and lipid accumulation of preadipocytes in the liver and regulate the decrease of TG content in the liver caused by exposure. Therefore, PS NPs and aPS NPs may promote the decomposition of triglyceride in the liver of mice by increasing the expression of ATGL, while the mouse self-regulation mechanism promotes the accumulation of lipids through up-regulating the expression of SREBP-1 and PPAR γ .

Glycogen is the building block of energy storage and structure in organisms, and high levels of blood glucose suggest possible mitochondrial dysfunction. PI3K activates the downstream molecule AKT by catalyzing the production of the second messenger phosphatidylinositol 3,4,5-triphosphate, while p-AKT activation further activates the downstream signaling molecule GLUT4, which is the terminal signal of intracellular insulin signaling that induces glucose uptake and intracellular glycogen synthesis in adipose and skeletal muscle tissue, and inhibits gluconeogenesis and glucose production in the liver. Interestingly, this phenomenon is the opposite of that observed in long term exposure in mice (Fan et al., 2022). A possible reason is that short term exposure to PS NPs and aPS NPs activates the blood glucose regulation system of the mouse so that the liver increases the expression levels of PI3K, p-AKT, and GLUT4 proteins to stimulate insulin secretion and promote hepatic glycogen synthesis, to inhibit excessive glucose production, thereby reducing blood sugar levels in mice. Long term exposure can lead to disturbances and even breakdowns of regulatory systems, eventually leading to diabetes in mice.

5. Conclusion

Collectively, we assessed the effects of short term exposure to PS NPs and aPS NPs on hepatic glucolipid metabolism in mice by analyzing pathology, blood biochemical markers, and metabolic profiles (metabolomics). The main results were as follows. First, exposure to both PS NPs and aPS NPs resulted in abnormal hepatic glucose and lipid metabolism. Second, compared with the PS NPs group, the glucose metabolism function of the mice in the aPS NPs exposed group was more significantly affected, which was also confirmed by the pathology and metabolomic results. Third, changes in serum LDL-C and HDL-C levels, increases in metabolites related to antioxidant defense and immune regulation, and up-regulation of related protein expression all indicate that the early injury caused by PS NPs and aPS NPs stimulation activates the self-defense mechanism in the body of the mouse. These findings not only provide the basis for the preliminary study of PS NPs and aPS NPs hepatotoxicity in mammals but also provide a reference for the selection of test substances for further research on the toxicity of NPs.

CRedit authorship contribution statement

Shiyu He: Conceptualization, Methodology, Software, Investigation, Formal analysis, Writing – original draft. **Jingran Wang:** Data curation, Experimental operation. **Lihong Zhou, Tianjiang Jia:** Visualization, Investigation. **Zhen Mao, Lingyan Zhang:** Resources, Supervision. **Xiaodan Zhang, Ji Wang, Man Yang:** Visualization, Writing – review & editing. **Peili Huang:** Conceptualization, Funding acquisition, Resources, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ecoenv.2023.114906.

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